

Numerical Methods for Solving the Wave Equation

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Quantum Numerical Methods

- Quantum methods involve solving the Schrodinger wave equation

$$H\psi = W\psi$$

- There are three common solution methods
 - Ab Initio – Most accurate, slowest
 - Density Functional – Less accurate, faster
 - Semi-Empirical – Least accurate, fastest

Demonstration

- The details of the three approaches are beyond the scope of these courses.
- However, we can demonstrate some capabilities using a simple commercial code named Hyperchem.